

Lecture 4 / Week 4

Roots of non-linear functioni) Bisection MethodStep I $\rightarrow$ Choose initial Interval

$$\begin{array}{c} [a \\ \downarrow \\ f(a) \\ -ve \end{array}$$

$$\begin{array}{c} b \\ \downarrow \\ f(b) \\ +ve \end{array}$$

Step II $\rightarrow$ (Assume root) $\rightarrow$

$$c = \frac{a+b}{2}$$

$$\begin{array}{c} \downarrow \\ f(c) \\ -ve \end{array}$$

$$\begin{array}{c} [c=a \\ \downarrow \\ f(c)=f(a) \\ -ve \end{array}$$

$$\begin{array}{c} b \\ \downarrow \\ f(b) \\ +ve \end{array}$$

$$\begin{array}{c} \downarrow \\ c' \\ \downarrow \\ f(c') \\ +ve \end{array}$$

$$\begin{array}{c} [a \\ \downarrow \\ f(a) \\ -ve \end{array}$$

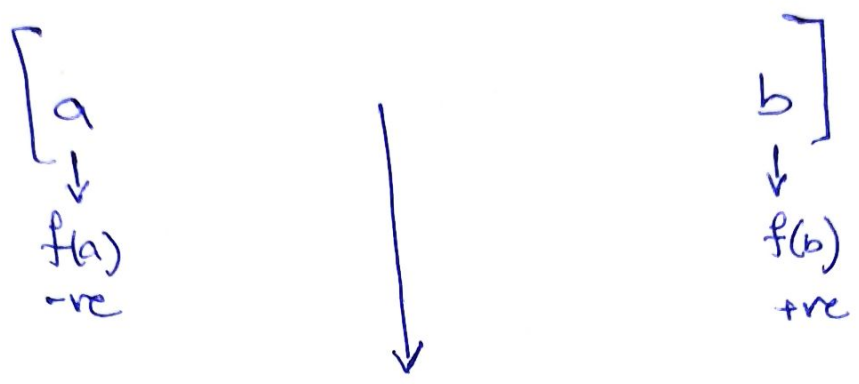
$$\begin{array}{c} [c=b \\ \downarrow \\ f(c)=f(b) \\ +ve \end{array}$$

$$\begin{array}{c} c'' \\ \downarrow \\ f(c'') \\ -ve \end{array}$$

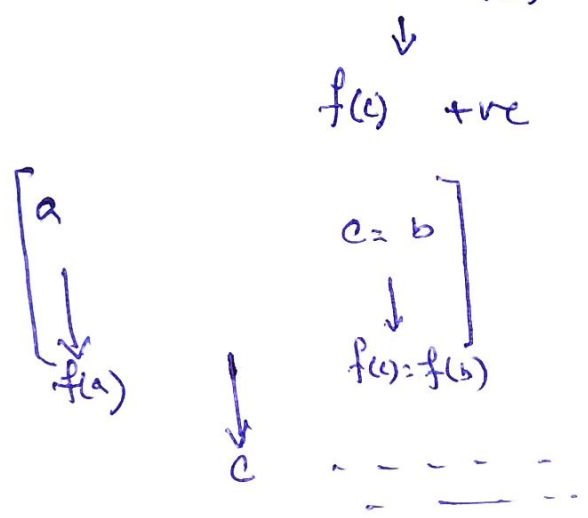
$$\begin{array}{c} [\\ \downarrow \\ \end{array}$$

ii) Regula Falsi Method

Step I Assume Initial Interval

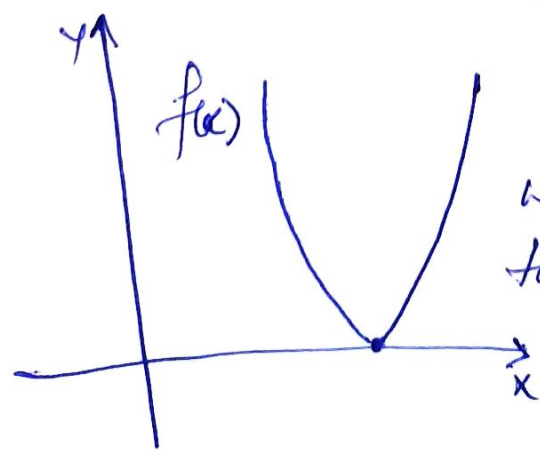


Step II $\rightarrow$ (Assume root) $\rightarrow C = a - \frac{b-a}{f(b)-f(a)} f(a)$



Short coming

Two initial inputs



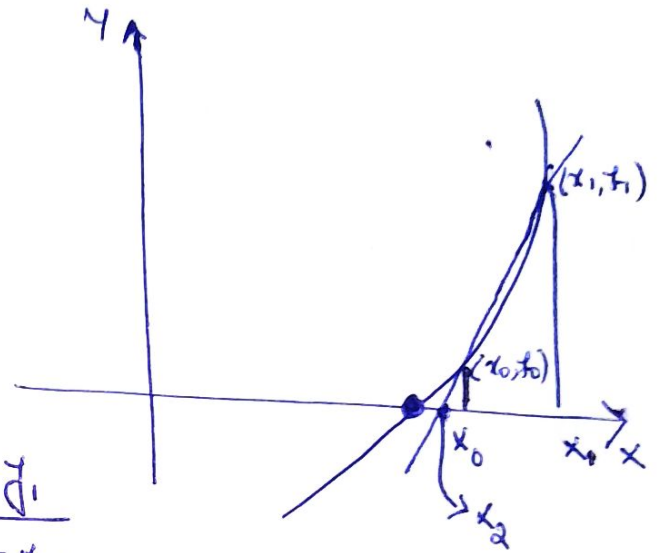
Cannot find the root where graph of $f(x)$ only touches the x-axis.

ii) Secant Method

Step I $\rightarrow$ choose any two points

$$(x_0, f(x_0)) \quad (x_1, f(x_1))$$

$$(x_0, f_0) \quad (x_1, f_1)$$



Eq. of secant line

$$y - y_1 = m(x - x_1) \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\boxed{y - f_0 = \frac{f_1 - f_0}{x_1 - x_0} (x - x_0)}$$

The point where secant line cuts the x-axis

$$y = 0 \quad x = x_2$$

$$0 - f_0 = \frac{f_1 - f_0}{x_1 - x_0} (x_2 - x_0)$$

$$\boxed{x_2 = x_0 - \frac{x_1 - x_0}{f_1 - f_0} f_0} \quad \text{and } f(x_2) = f_2 = \dots$$

Now ignore first point and take

$$(x_1, f_1) \quad (x_2, f_2)$$

Eq. of secant line

$$y - y_1 = m(x - x_1)$$

$$\boxed{y - f_1 = \frac{f_2 - f_1}{x_2 - x_1} (x - x_1)}$$

The point where line cuts the x-axis

$$y = 0 \quad x = x_3$$

$$0 - f_1 = \frac{f_2 - f_1}{x_2 - x_1} (x_3 - x_1)$$

$$\boxed{x_3 = x_1 - \frac{x_2 - x_1}{f_2 - f_1} f_1}$$

$$f(x_3) = f_3 = \dots$$

Now new points are

$$(x_2, f_2) \quad (x_3, f_3)$$

$$\boxed{x_4 = x_2 - \frac{x_3 - x_2}{f_3 - f_2} f_2}$$

⋮

$$\boxed{x_{n+1} = x_{n-1} - \frac{x_n - x_{n-1}}{f_n - f_{n-1}} f_{n-1}}$$

Formula.

Example ①

Starting with $x_0=1$ and $x_1=2$ apply Regula-Falsi method on the function $f(x)=x^2+4x-10$ to obtain an approximate root correct to 4-D places.

Solution

$$f(x) = x^2 + 4x - 10$$

$$x_0 = 1$$

$$x_1 = 2$$

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f_n - f_{n-1}} f_n$$

$$f_0 = -5$$

$$f_1 = 2$$

n	x_{n-1}	$f(x_{n-1})$	x_n	$f(x_n)$	x_{n+1}	$f(x_{n+1})$
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1	$x_0 = 1$	$f(x_0) = f_0 = -5$	$x_1 = 2$	$f(x_1) = f_1 = 2$	$x_2 = x_0 - \frac{x_1 - x_0}{f_1 - f_0} f_0$ $x_2 = 1 - \frac{(2-1)(-5)}{(2-(-5))} = -0.204080$ $= 1.714286$	$f(x_2) = f_2$
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2	$x_1 = 2$	$f_1 = 2$	$x_2 = 1.714286$	$f_2 = -0.204080$	$x_3 = x_1 - \frac{x_2 - x_1}{f_2 - f_1} f_1$ $= 1.740741$	-0.006857
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3	$x_2 = 1.714286$	$f_2 = -0.204080$	$x_3 = 1.740741$	$f_3 = -0.006857$	$x_4 = 1.741627$	-0.000227
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4	$x_3 = 1.740741$	$f_3 = -0.006857$	$x_4 = 1.741627$	$f_4 = -0.000227$	$x_5 = 1.741656$	-0.000010
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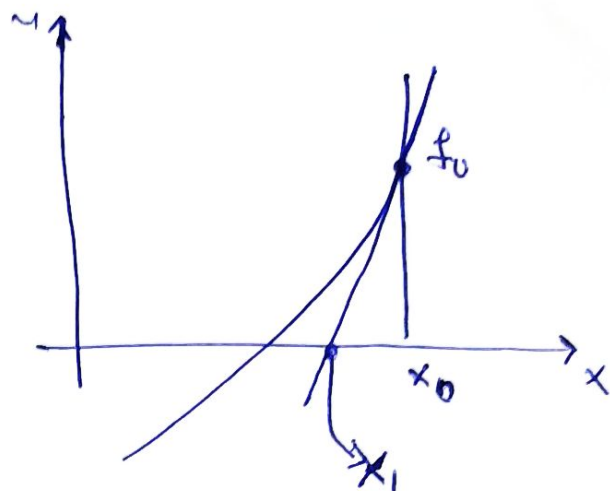
Ans.

iv) Newton's Method

Choose one point

$$(x_0, f(x_0))$$

$$(x_0, f_0)$$



Eq. of tangent line

$$y - y_1 = m(x - x_1)$$

$$m = f'(a)$$

$$y - f_0 = f'(x_0)(x - x_0)$$

The point where tangent line cuts the x-axis

$$y = 0 \quad x = x_1$$

$$0 - f_0 = f'(x_0)(x_1 - x_0)$$

$$x_1 = x_0 - \frac{f_0}{f'(x_0)}$$

$$f(x_1) = f_1 = \dots$$

Now new point (x_1, f_1)

$$y - y_1 = m(x - x_1)$$

$$y - f_1 = f'(x_1)(x - x_1)$$

The point where line cuts the x-axis

$$y = 0 \quad x = x_2$$

$$0 - f_1 = f'(x_1)(x_2 - x_1)$$

$$\boxed{x_2 = x_1 - \frac{f_1}{f'(x_1)}} \rightarrow f(x_2) = f_2 = \dots$$

New point (x_2, f_2)

$$\boxed{x_3 = x_2 - \frac{f_2}{f'(x_2)}}$$

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

Formula.

(8)

Example

Starting with $x_0 = 1$ approximate the root of the function $f(x) = x^2 + 4x - 10$ correct to 6-D places.

Solution

$$f(x) = x^2 + 4x - 10$$

$$x_0 = 1$$

$$f'(x) = 2x + 4$$

$$f(x_0) = f_0 = -5$$

$$x_n = x_{n-1} - \frac{f_{n-1}}{f'_{n-1}}$$

n	x_{n-1}	$f(x_{n-1})$	$f'(x_{n-1})$	x_n	$f(x_n)$
1	$x_0 = 1$	$f(x_0) = f_0 = -5$	$f'(x_0) = f'(1) = 6$	$x_1 = x_0 - \frac{f_0}{f'_0}$ $= 1 - \frac{(-5)}{6}$ $= 1.8333333$	$f(x_1) = 0.6944442$
2	$x_1 = 1.8333333$	$f_1 = 0.6944442$	$f'(x_1) = 7.6666667$	$x_2 = \frac{1.7427536}{1.7427536}$	0.0082045
3	$x_2 = 1.7427536$	$f_2 = 0.0082045$	$f'(x_2) = 7.4855072$	$x_3 = 1.7416575$	0.0000000 $0.62 \dots$

